

50 MCQs, 30 short, 20 long, 40 viva, 12 cases, 10 data questions - with the full instructor answer key

Instructor assessment package

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All datasets used in this package are simulated teaching data. Wet-lab and animal procedures described here are conceptual; institutional biosafety approval and SOPs govern all practice.

Table of Contents

Placeholder for table of contents	0
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Assessment — 50 Multiple-Choice Questions

Course: GMOs, Cloning Techniques, and Developmental & Spatial Gene Expression

Levels: Recall → Understanding → Application → Analysis → Evaluation

Answers & explanations: Answer-Key.md

Section A — Recall (Q1–10)

1. Which enzyme seals phosphodiester bonds during cloning?
A) Restriction endonuclease · B) DNA ligase · C) DNA polymerase · D) Reverse transcriptase
2. The sequence EcoRI recognizes is:
A) G^AAATTC · B) GGATCC · C) AAGCTT · D) CTGCAG
3. A plasmid origin of replication controls:
A) Insert size limit · B) Copy number · C) Promoter strength · D) Antibiotic resistance
4. GFP was originally isolated from:
A) *Discosoma* sp. · B) Firefly · C) *Aequorea victoria* · D) *Renilla reniformis*
5. The PAM for SpCas9 is:
A) 5'-NGG-3' · B) 5'-TTTV-3' · C) 5'-NNGRRT-3' · D) 5'-NG-3'
6. The "T" in a TA-cloning T-vector pairs with:
A) A 5'-A overhang · B) A 3'-A overhang · C) A blunt end · D) An RNA cap
7. Which model organism has an invariant cell lineage?
A) Mouse · B) Zebrafish · C) *C. elegans* · D) *Arabidopsis*
8. Blue-white screening uses disruption of:
A) ampR · B) lacZ α · C) ori · D) GFP
9. Dolly the sheep was produced by:
A) Pronuclear microinjection · B) Somatic-cell nuclear transfer · C) ES-cell complementation · D) Viral transduction
10. Colinearity refers to the alignment between Hox gene order and:
A) Chromosome size · B) Expression along the body axis · C) GC content · D) Replication timing

Section B — Understanding (Q11–20)

11. Dephosphorylating the vector before ligation primarily prevents:
A) Insert concatemerization · B) Vector self-ligation · C) Star activity · D) Methylation
12. Type IIS enzymes (e.g., BsaI) are useful in Golden Gate because they:
A) Cut only methylated DNA · B) Cut outside their recognition site · C) Leave blunt ends · D) Require RNA cofactors
13. A reporter gene under a tissue-specific promoter reports:
A) Protein folding · B) Where the promoter is active · C) Genome position effects · D) mRNA half-life
14. In RT-qPCR, the $\Delta\Delta C_t$ method assumes:
A) 100% primer efficiency · B) Single-cell input · C) RNA is double-stranded · D) Reference genes vary by condition
15. NHEJ repair of a Cas9 cut usually produces:

- A) Precise donor integration · B) Small indels · C) Base conversion A→G · D) methylation changes
- 16.** An enhancer, compared with a promoter, is characteristically:
- A) Position- and orientation-flexible · B) Always adjacent to the TSS · C) Coding · D) Prokaryotic-only
- 17.** Gibson Assembly requires fragments with:
- A) Blunt ends · B) Terminal sequence overlaps · C) 3'-A overhangs · D) Identical restriction sites
- 18.** Single-cell RNA-seq primarily loses:
- A) Gene identities · B) Spatial position of cells · C) UMI counts · D) Cluster structure
- 19.** In spatial transcriptomics (array-based), one capture spot typically integrates:
- A) One chromosome · B) Several cells · C) One nucleus · D) A whole organ
- 20.** A BAC differs from a standard plasmid chiefly in:
- A) Antibiotic marker · B) Insert capacity · C) Circular topology · D) Use of DNA ligase

Section C — Application (Q21–32)

- 21.** You must clone a 2 kb insert with an internal BamHI site into a vector; the best direct strategy is:
- A) BamHI single digest · B) Two other compatible unique sites flanking the MCS · C) Ligase-only blunt ligation with no preparation · D) Skip validation
- 22.** A colony is PCR-positive with vector+insert primers but the diagnostic digest disagrees with the map. You should trust:
- A) The PCR — it is more sensitive · B) The digest — structural evidence · C) Coin-flip · D) Discard without analysis
- 23.** To quantify promoter activity over time in cell culture with the widest dynamic range, choose:
- A) lacZ/X-gal · B) Firefly luciferase (normalized) · C) Colony PCR · D) Sanger sequencing
- 24.** For *absolute* transcript counts per single cell in tissue, choose:
- A) Bulk RNA-seq · B) smFISH · C) Western blot · D) Colony PCR
- 25.** A zebrafish F0 crispant shows mixed phenotypes across tissues; most likely explanation:
- A) Sequencing error · B) Mosaicism · C) HDR repair · D) Sense-probe signal
- 26.** To test whether a candidate enhancer is *necessary* for endogenous expression, the best first experiment is:
- A) Enhancer-reporter assay · B) CRISPRi/knockout of the endogenous enhancer + ISH readout · C) Western blot of the enhancer · D) Delete the promoter
- 27.** An RT-qPCR shows *myod1* up 8-fold at 18 hpf; the strongest independent spatial validation would be:
- A) More qPCR replicates · B) ISH showing somite/paraxial signal at 18 hpf · C) Genomic PCR · D) Colony PCR
- 28.** In a ligation with insert:vector 0:1 (vector only, not dephosphorylated), expected colony outcome:
- A) Zero colonies · B) Background empty-vector colonies · C) All correct clones · D) Lambda phage plaques
- 29.** Your Gibson assembly uses 12 bp overlaps between fragments; likely result:
- A) Improved fidelity · B) Inefficient/unreliable assembly · C) Faster reaction · D) Star activity
- 30.** To generate a C-terminal GFP fusion, the insert design must:
- A) Include a stop codon before GFP · B) Lack a stop codon and be in-frame with GFP · C) Be blunt-ended · D) Use T7 terminator
- 31.** A KO experiment yielded a viable F0 whose target PCR shows a 3-bp in-frame deletion; interpretation:

A) Knockout succeeded · B) Protein function may be preserved — screen for frameshift alleles or use protein-level validation · C) HDR occurred · D) The guide bound RNA

32. Which design change best reduces off-target risk?

A) Longer ligation · B) Guide re-design with off-target scoring + RNP delivery · C) More antibiotic · D) Higher-copy vector

Section D — Analysis (Q33–42)

33. A digest of a "correct" construct yields fragments summing to 5.8 kb, but the map says 4.05 kb. Best conclusion:

A) Pippetting error only · B) Rearranged/duplicated plasmid · C) Perfect match · D) Sense-probe contamination

34. A DE analysis at $FDR < 0.05$ returns 1,800 genes with $|\log_2 FC| < 0.1$. Most likely explanation:

A) Biology is subtle but real · B) Under-powered/near-zero variance artifact — reconsider normalization/design · C) Genes are misnamed · D) FDR too strict

35. In a scRNA-seq UMAP, one cluster co-expresses *pax6a* and *myod1* at low levels. Most defensible interpretation:

A) Sequencing artifact — ignore · B) Possible transition state or doublet — verify with additional markers/QC · C) New species · D) Sense probe

36. A spatial matrix shows gene X high at the tissue edge, declining inward ($r = -0.6$ vs distance). This is best described as:

A) A domain · B) A gradient · C) A mosaic · D) A mutation

37. An enhancer-reporter works in orientation A but not B, reproducibly. Defensible statement:

A) It is a classic enhancer · B) It behaves promoter-like/orientation-dependent — check for cryptic promoter activity · C) Enhancers never show orientation effects · D) The Renilla failed

38. In a time course, a gene starts high and decays over early stages. Best annotation:

A) Zygotic activation · B) Maternal contribution · C) Housekeeping · D) Reporter artifact

39. A heat-killed "positive control" for transformation produces 300 colonies; your ligation plate has 2. Strongest conclusion:

A) Competent cells are fine — ligation/assembly failed · B) Cells are dead · C) Antibiotic wrong · D) Transformation worked great

40. Colony PCR (Set A, flanking primers) cannot determine orientation because:

A) Primers are identical on both strands · B) Product size is the same for both orientations · C) DNA is circular · D) PCR can't amplify plasmids

41. In an RNA-seq PCA, samples separate by batch, not condition. Correct action:

A) Report condition results anyway · B) Re-balance design or model the batch; do not interpret condition from this run · C) Delete batch samples · D) Increase reads 10×

42. An ISH sense-probe panel shows moderate uniform signal. Correct interpretation:

A) Gene is expressed everywhere · B) Background/trapping — antisense comparisons must subtract this · C) Probe degraded · D) Stage mismatch only

Section E — Evaluation (Q43–50)

43. Which claim does a promoter-GFP zebrafish line NOT support?

A) The promoter is active in GFP-positive cells · B) GFP pattern matches promoter activity · C) The endogenous gene is expressed identically · D) The construct works in vivo

44. Golden Rice's primary scientific claim rests on:

A) Higher yield · B) Provitamin A accumulation in endosperm · C) Herbicide tolerance · D) Nitrogen fixation

45. "GMOs are 100% safe" as a universal statement is scientifically problematic because:

A) Safety is assessed per product/trait, not by category · B) GMOs don't exist · C) Safety cannot be studied · D) All assessments failed

46. Which pair correctly matches method and question?

A) ISH — protein localization · B) IF — mRNA localization · C) smFISH — absolute mRNA counts · D) qPCR — spatial maps

47. A reviewer asks whether your CRISPRi enhancer result could be off-target seed effects. Best response:

A) Ignore · B) Include independent sgRNAs and a non-targeting control ± rescue/motif-mutant comparison · C) Lower the FDR · D) Use more antibiotic

48. Bt-corn resistance evolution in target pests is best addressed by:

A) Abandoning the trait · B) Refuge + pyramided-toxin management strategies · C) Higher insecticide spraying · D) Ignoring monitoring

49. Which statement about germline editing of humans is most defensible in a scientific-ethics exam?

A) It is already routine · B) Technical possibility does not equal ethical readiness; oversight frameworks (e.g., WHO 2021) call for strict limits · C) No risks exist · D) Consent is unnecessary for future generations

50. The best final validation before declaring a new construct "confirmed":

A) Colony count · B) Sanger sequencing across junctions and full insert · C) Blue colony color · D) Correct antibiotic growth

Total: 50 questions. Answer key with explanations: Answer-Key.md · Also see Short Questions, Long Questions, Viva Questions, Case Studies

Assessment — 30 Short-Answer Questions

Expected length: 3–6 sentences each · **Model answers:** Answer-Key.md

1. Define recombinant DNA technology and name the landmark 1973 experiment that established it.
2. Distinguish transgenic from cisgenic organisms with one example each.
3. Why is an origin of replication essential in a plasmid vector?
4. State the role of three components of an expression cassette.
5. What is a multiple cloning site and why is it useful?
6. Define directional cloning and name the design feature that achieves it.
7. Why is vector dephosphorylation used in ligation setups?
8. Give one advantage and one limitation of TA cloning.
9. What makes Type IIS enzymes central to Golden Gate assembly?
10. Describe the three enzymatic activities in a Gibson Assembly reaction.
11. Explain the difference between selection and screening with one example each.
12. How does blue-white screening distinguish empty from insert-carrying colonies?
13. Why is a colony on a selective plate only a *candidate* clone?
14. What does Sanger sequencing add beyond a correct restriction digest?
15. Distinguish constitutive, inducible, and tissue-specific promoters.
16. Why does GFP require no exogenous substrate?
17. When is luciferase preferable to GFP as a reporter, and why?
18. Define morphogen and state the French-flag model in one sentence.
19. What is colinearity in Hox gene organization?
20. Name two signaling pathways that pattern the vertebrate neural tube and their sources.
21. Define cell determination and give an experimental criterion for it.
22. Distinguish spatial from temporal gene expression with a concrete example of each.
23. Why does bulk RNA-seq average away spatial information?
24. What is the purpose of the sense probe in ISH?
25. Why must reference genes be validated for each qPCR experiment?
26. State the $\Delta\Delta C_t$ formula and the assumption behind it.
27. What is mosaicism in F0 genome editing and why does it matter?
28. Give one example each of NHEJ- and HDR-based outcomes in CRISPR editing.
29. Why is dropout a caution when interpreting scRNA-seq absence calls?
30. State the four layers of GMO risk reasoning in the correct order.

Grading guide: each question is scored for (1) correct concept, (2) precision of terminology, (3) example or criterion where demanded. Model answers in Answer-Key.md.

Assessment — 20 Long-Answer Questions

Expected length: 1–2 pages with diagrams where useful · **Model answer outlines:** Answer-Key.md

Molecular cloning & vectors

1. Describe the complete molecular-cloning workflow from target-gene identification to sequence-confirmed construct. For each stage, state purpose, common errors, and validation.
2. Compare restriction–ligation, Gibson, Golden Gate, and Gateway cloning. Include mechanism, strengths, limitations, and a justified scenario for each.
3. Explain plasmid vector architecture in detail: for each component (ori, marker, MCS, promoter, terminator, reporter), state its function and the consequence of its absence.
4. Design (on paper) an expression construct for a zebrafish developmental gene with a C-terminal His tag: elements, order, cloning strategy, screening, and validation plan.
5. Discuss expression-vector design choices that control expression level, timing, and tissue specificity, with one concrete example of each change and its predicted effect.

GMO generation & CRISPR

6. Compare the conceptual workflows for generating genetically modified bacteria, plants, and animals: DNA delivery, selection, regeneration/development, and validation. Include one landmark example per kingdom.
7. Explain Agrobacterium-mediated plant transformation at the molecular level: Ti plasmid, T-DNA borders, vir functions, binary vectors, and why totipotency matters.
8. Describe the CRISPR–Cas9 mechanism from guide-RNA design through target recognition to repair-pathway choice, and explain how knockout vs knock-in outcomes arise.
9. Compare base editing and prime editing with nuclease editing: mechanisms, what edits each achieves, and the trade-offs.
10. Critically compare traditional transgenesis with genome editing: what each is best for, and why many modern projects use both. Include off-target and mosaicism considerations.

Developmental & spatial gene expression

11. Explain differential gene expression as the basis of development. Include cell fate, determination, differentiation, potency, and the master-regulator concept with its caveat.
12. Describe the morphogen concept using Shh in the vertebrate neural tube: source, gradient, thresholds, fates, and why duration of exposure matters.
13. Outline the Drosophila segmentation GRN hierarchy (maternal → gap → pair-rule → segment-polarity → Hox), explaining how each level refines spatial precision.
14. Compare enhancers, silencers, promoters, and operators: location, activity logic, and experimental demonstration methods for each.
15. Explain gene-regulatory-network motifs (positive feedback, mutual repression, feed-forward, lateral inhibition) and the developmental behavior each produces, with one example each.

Methods & data analysis

16. Compare RT-qPCR, RNA-seq, scRNA-seq, and spatial transcriptomics: question answered, workflow, output, and the interpretation each supports. Include a decision guide for choosing among them.
17. Describe the in-situ-hybridization workflow end to end, including probe logic (antisense/sense), controls, and how you would interpret a domain vs a gradient from ISH data.
18. A qPCR experiment reports an 8-fold increase of *myod1* at 18 hpf. Design the full validation and extension plan: statistical treatment, spatial validation, protein-level check, and functional follow-up.

19. Explain the single-cell RNA-seq computational workflow (QC → normalization → PCA → clustering → annotation → trajectory) and the biological claim each step licenses.
 20. Discuss reporter-based expression analysis: construct types (promoter-, enhancer-, fusion-reporter), controls, interpretation limits, and how reporter data should be validated against endogenous expression.
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Grading guide (per question, 10 pts): scientific accuracy (4) · completeness vs the prompt (3) · correct use of examples/evidence (2) · clarity and structure incl. diagrams (1). Model outlines in Answer-Key.md.

Assessment — 40 Viva-Voce Questions

Format: rapid oral exam; student answers in 1–3 sentences; examiner probes with follow-ups (listed where useful). **Answer guidance:** Answer-Key.md

Cloning & vectors (1–10)

1. What is the difference between an insert and a vector?
2. Name three functions carried by a plasmid backbone.
3. Why does copy number matter when expressing a toxic gene?
4. What ends does EcoRI leave, and why are they called "sticky"?
5. Why can BamHI and BglII ends ligate to each other?
6. What does T4 DNA ligase need as a cofactor?
7. Why is a 3:1 insert:vector ratio a starting point rather than a law?
8. In blue-white screening, what does a white colony tell you — and what doesn't it?
9. Why is colony PCR with one vector + one insert primer orientation-specific?
10. What is a shuttle vector — give a concrete use case.

Techniques (11–16)

11. Why does Gibson Assembly not need restriction sites?
12. What enzyme property makes Golden Gate overhangs "programmable"?
13. Why does a Golden Gate reaction become more product-specific as it proceeds?
14. What is an entry clone in Gateway-style logic?
15. When would blunt-end cloning be your only option?
16. Why must high-fidelity PCR products be A-tailed before TA cloning?

GMO & CRISPR (17–26)

17. Define transgenic organism in one sentence.
18. Why is Dolly the sheep not a transgenic animal?
19. What is a binary vector system in plant transformation?
20. Why does totipotency make plant engineering conceptually simpler?
21. What is a PAM and why can't Cas9 cut without one?
22. NHEJ vs HDR — one-line definitions.
23. Why do knock-ins require more design than knockouts?
24. What is mosaicism and when does it arise?
25. What is the difference between CRISPRi and CRISPRa?
26. Why do base editors avoid the HDR-efficiency problem?

Reporters & expression (27–33)

27. Why does GFP not need a substrate?
28. When would you pick lacZ over GFP?
29. What does the Renilla control correct for in a luciferase assay?
30. What is a destabilized reporter and what question does it answer?
31. Why can a reporter pattern disagree with endogenous ISH?
32. What is a Kozak sequence?
33. What changes when you swap a constitutive promoter for a tissue-specific one?

Developmental & spatial expression (34–40)

34. Define differential gene expression.
 35. What is a morphogen gradient — give one real example.
 36. What are Hox genes and what does colinearity mean?
 37. Gradient vs domain — how would you distinguish them quantitatively?
 38. Why does scRNA-seq lose spatial information, and which method restores it?
 39. What does a sense-probe control measure in ISH?
 40. Why do statistics on ΔC_t rather than fold change in qPCR?
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Examiner follow-up bank (use to probe depth):

- "Give an example." · "What would happen if that component were missing?" · "Which control would detect that failure?" · "State the limitation of that method." · "How would you validate that claim independently?"

Assessment — 12 Case-Based Questions + 10 Experimental/Data-Interpretation Questions

Model answers: Answer-Key.md · Background reading: Case-Studies module

Part I — Case-based questions (12)

Case 1 — The failing Bt field. A region growing Bt cotton for 8 years reports rising insect damage.

(a) What evolutionary process explains this? (b) Why do refuge strategies slow it? (c) Propose two monitoring data streams a program should collect.

Case 2 — Golden Rice approval. A country must decide on GR2E deployment.

(a) Summarize the mechanism. (b) What food-safety evidence would you require? (c) Separate the scientific, ethical, and policy layers of the decision (Module 14 framework).

Case 3 — Insulin to the clinic. Explain why *E. coli* was viable for insulin production but not for a glycoprotein like EPO, and what host features each requires.

Case 4 — The GFP that vanished. A student's promoter-GFP line shows bright signal at F0 but fading lines over generations.

(a) List three mechanisms (silencing, position effects, breeding-away). (b) Which molecular confirmation distinguishes them?

Case 5 — A confusing disease model. An *apoE* mouse model shows diet-dependent lesions but a new drug fails in trial.

(a) What does this teach about model validity? (b) Name the three validity types used in modeling.

Case 6 — The in-frame escape. A CRISPR knockout targeting exon 2 yields a founder with a 3-bp deletion, healthy phenotype.

(a) Why can this happen? (b) Design two follow-ups to establish whether function is truly lost.

Case 7 — Edited mushroom controversy. A non-browning mushroom produced by CRISPR knockout historically escaped USDA regulation in the US.

(a) Why (process vs product framing)? (b) Contrast with an EU-style process-based assessment outcome.

Case 8 — Antennapedia. A fly misexpresses a Hox gene and grows legs where antennae belong.

(a) What does this demonstrate about Hox function? (b) Why does colinearity suggest regulatory/chromatin mechanisms?

Case 9 — Bicoid rescue. A *bicoid* null mother produces headless embryos; anterior *bicoid* mRNA injection rescues patterning.

(a) What does the rescue prove about Bicoid's role? (b) What does the gradient's formation tell you about maternal vs zygotic contribution?

Case 10 — The Shh dose question. Ectopic Shh in the dorsal neural tube induces floor-plate markers near the source and motor neurons further away.

(a) What model explains the two fates? (b) What experiment separates concentration from duration effects?

Case 11 — Lost coordinates. A scRNA-seq of developing gut finds an "absorptive precursor" cluster; a reviewer asks where these cells are.

(a) Why can't scRNA-seq answer this? (b) Name two methods that would, and what each adds.

Case 12 — Spatial rumor. A spatial-transcriptomics map suggests ligand L in region A neighbors receptor R in region B.

(a) What inference does this license? (b) What validation would upgrade it from hypothesis to finding?

Part II — Experimental/data-interpretation questions (10)

(Datasets referenced are in `DATA`; all simulated.)

D1 — Gel. A digest of candidate clone X shows bands at 3.2 kb only; clone Y at 4.05 + 0.85 kb. Which is correct, and what is X? (Dataset: cloning-data/)

D2 — Fragment sum. Clone Z's digest gives 2.3 + 1.7 + 0.9 kb; the map says the plasmid is 3.2 kb. Diagnose with reasoning.

D3 — Colony PCR table. Using `colony_pcr_results.csv`, tabulate how many of the 24 colonies are correct / empty / wrong-orientation / ambiguous, and justify one discrepancy vs plate color.

D4 — Transformation counts. From `transformation_plates.csv`, rank conditions L1–L4 by *correct-clone* rate (not colony count) and explain why the rankings differ.

D5 — qPCR. From `qPCR_timecourse.csv`, compute *myod1* fold change at 18 hpf vs 10 hpf (geometric-mean reference) and state the statistical treatment you would report.

D6 — Reference failure. Your qPCR shows apparent 3-fold *shha* induction using *ef1a* alone, but the stability table flags *ef1a* as stage-variable. Recompute with the stable reference and reinterpret.

D7 — Volcano. From `rnaseq_counts.csv`, produce (or describe) the volcano plot; how many genes pass $FDR < 0.05$ AND $|\log_2 FC| > 1$, and what module do the top hits form?

D8 — Clusters. From `single_cell_counts.csv`, cluster at $k=3$ and annotate each cluster with its marker gene. What does the $k=4$ subcluster suggest?

D9 — Spatial marker. From `spatial_matrix.csv` + `spot_coordinates.csv`, identify the top marker of each region and plot one marker's x,y distribution. Which region would you call "notochord-like" and why?

D10 — Gradient test. Correlate the edge-gradient gene's expression with distance-to-edge ($r \approx ?$). State the criterion that distinguishes this pattern from a domain, and the ISH experiment that would validate it.

Grading guide: case questions are scored for mechanism accuracy (40%), layering of evidence/risk/ethics/policy where relevant (30%), and proposed experiments' validity (30%). Data questions: correct interpretation (50%), correct quantitative handling (30%), stated limitations (20%).

Instructor Answer Key

Covers: MCQs · Short Questions · Long Questions · Viva Questions · Case Studies

Instructor-only note: issue to students only after assessment, or use sections selectively.

Part 1 — MCQ answers (with explanations)

Q	Ans	Explanation
1	B	DNA ligase seals 5'P–3'OH nicks; restriction enzymes cut.
2	A	EcoRI: G ^A AATTC (sticky 5'-AATT).
3	B	ori sets replication initiation → copy number.
4	C	GFP from <i>Aequorea victoria</i> ; RFP from <i>Discosoma</i> ; luciferases from firefly/ <i>Renilla</i> .
5	A	SpCas9 PAM = NGG; TTV is Cas12a; NNGRRT is SaCas9.
6	B	Taq adds non-templated 3'-A; T-vector provides 3'-T.
7	C	<i>C. elegans</i> lineage is invariant (Sulston).
8	B	Insertion disrupts lacZ α → white colonies.
9	B	Dolly = SCNT (Wilmut et al. 1997).
10	B	Hox order matches anterior→posterior expression.
11	B	No 5' phosphates on vector → cannot self-ligate.
12	B	Type IIS cuts outside the recognition site → designer overhangs.
13	B	Reporter output maps promoter activity spatially.
14	A	$\Delta\Delta C_t$ assumes ~100% efficiency; otherwise use efficiency-corrected methods.
15	B	NHEJ → small indels → frameshift/knockout.
16	A	Enhancers act at variable distances/orientations.
17	B	Gibson needs 20–40 bp terminal overlaps.
18	B	Dissociation removes spatial coordinates.
19	B	Spots ≈ several cells (deconvolution needed).
20	B	BACs carry 100–300 kb via F-factor ori.
21	B	Use two unique sites that avoid the internal site; keep directionality.
22	B	Digest = structural evidence; PCR can amplify through anomalies.
23	B	Luciferase: widest dynamic range + normalization.
24	B	smFISH gives absolute per-cell counts.

Q	Ans	Explanation
25	B	Post-one-cell editing → mosaicism.
26	B	Necessity requires endogenous perturbation; reporter tests sufficiency.
27	B	ISH validates spatial pattern at the same stage.
28	B	Vector-only → background empty-vector colonies.
29	B	12 bp overlaps are too short; use 20–40 bp.
30	B	C-terminal fusion needs in-frame join, no premature stop.
31	B	In-frame deletions can preserve function; screen for frameshifts/validate protein.
32	B	Guide re-design + RNP reduces off-target exposure.
33	B	Fragment sum > map size → duplication/rearrangement.
34	B	Near-zero FCs at scale = design/variance artifact; re-examine.
35	B	Co-expression could be transition state or doublet — verify.
36	B	Continuous decline with distance = gradient.
37	B	Orientation-dependence suggests promoter-like behavior — investigate.
38	B	Early-high, decaying = maternal contribution.
39	A	Positive control worked → problem is upstream (ligation/assembly).
40	B	Flanking primers give equal product size either orientation.
41	B	Batch-dominant PCA invalidates condition readout — fix design/model.
42	B	Sense = background; subtract/compare, don't over-interpret.
43	C	Reporter ≠ endogenous expression claim.
44	B	Golden Rice = provitamin A in endosperm.
45	A	Safety is product/trait-specific, not category-wide.
46	C	smFISH = absolute counts; other pairings are mismatched.
47	B	Independent sgRNAs + controls address off-target concern.
48	B	Refuge + pyramids = resistance management.
49	B	WHO 2021 framework: oversight, strict limits; not routine.
50	B	Sanger across junctions + insert = definitive sequence confirmation.

Scoring: 2 pts each ($50 \times 2 = 100$). Level mapping: Q1–10 recall; Q11–20 understanding; Q21–32 application; Q33–42 analysis; Q43–50 evaluation.

Part 2 — Short-answer model answers (condensed)

1. **rDNA technology:** joining DNA from >1 source into a propagatable molecule; Cohen–Chang–Boyer–Helling 1973 (functional recombinant plasmids in *E. coli*).
2. **Transgenic** = foreign-species DNA (Bt cotton); **cisgenic** = same-species DNA (potato with potato resistance gene).
3. **ori:** replication initiation site; without it the plasmid is lost → no colonies.
4. **Cassette:** promoter (initiation), RBS/Kozak (translation initiation), terminator/polyA (defined end/stability).
5. **MCS:** cluster of unique sites → many enzyme choices in one region; directional cloning feasible.
6. **Directional cloning:** two different enzymes on vector + insert ends → only one orientation possible.
7. **Dephosphorylation:** removes vector 5'P → blocks self-ligation; insert supplies phosphates.
8. **TA:** + fast PCR capture; – no orientation control, incompatible with blunt (high-fidelity) products without A-tailing.
9. **Type IIS:** cut outside recognition site → overhang sequence is designer-chosen → unique addresses per junction.
10. **Gibson:** 5' exonuclease chew-back → overlap annealing → polymerase fill → ligase seal.
11. **Selection** = condition kills undesired cells (antibiotic); **screening** = per-colony test (colony PCR).
12. **Blue-white:** insert disrupts lacZα → no β-gal → white on X-gal; empty → blue.
13. **Candidate:** selection only proves "has vector"; insert presence/orientation/sequence unproven.
14. **Sanger:** actual nucleotide sequence — detects point mutations/frameshifts invisible to digests.
15. **Promoters:** constitutive = always on; inducible = signal-gated (Tet, lac); tissue-specific = spatial restriction.
16. **GFP:** autocatalytic chromophore (Ser65–Tyr66–Gly67), needs only O₂ → genetically encoded fluorescence.
17. **Luciferase:** quantitative dynamic range, plate-reader assays, dual normalization; weaker spatial detail.
18. **Morphogen/French-flag:** concentration gradient + thresholds → distinct fates at distinct distances.
19. **Colinearity:** Hox chromosomal order matches A–P expression order/timing.
20. **Neural tube:** Shh (notochord/floor plate, ventral); BMP (roof plate/ectoderm, dorsal).
21. **Determination:** fate fixed despite transplantation (classic organizer/somite experiments).
22. **Spatial/temporal:** *shh* in notochord (where); *myod1* rise at 12–18 hpf (when).
23. **Bulk RNA-seq:** averages whole tissue → a small hot region dilutes into the whole.
24. **Sense probe:** cannot pair with target → measures non-specific background/trapping.
25. **Reference validation:** housekeeping genes vary with stage/treatment → invalid normalization if unvalidated.
26. $\Delta\Delta Ct$: $\Delta Ct = Cq_t - Cq_{ref}$; $\Delta\Delta Ct = \Delta Ct_{cond} - \Delta Ct_{ctrl}$; $FC = 2^{(-\Delta\Delta Ct)}$; assumes ~100% efficiency.
27. **Mosaicism:** edits after first division → different alleles in different tissues → genotyping/phenotype complications.
28. **NHEJ outcome:** indels → knockout. **HDR outcome:** donor-templated correction/knock-in.
29. **Dropout:** low transcript undetected ≠ absent biology — absence claims need care.
30. **Four layers:** scientific evidence → risk assessment → ethical considerations → policy/regulation.

Part 3 — Long-question outlines (key points expected)

- L1.** Workflow stages with purpose/errors/validation per stage (Module 2 §2); must include screening→digest→sequencing logic.
 - L2.** Comparison table mechanism/strength/limitation + one justified scenario per method (Module 4 §7); scenario justification is the discriminator.
 - L3.** Each component's job + failure consequence (Module 2 §1 table); full marks require the "consequence if absent" column.
 - L4.** Tag/terminator order correct; frame logic; screening + sequencing plan; enzyme choices avoid internal sites.
 - L5.** Element swaps → predicted expression change; e.g., inducible promoter for toxic gene; tissue-specific promoter converts localization study.
 - L6.** Three kingdoms' workflows + one landmark each (insulin; Bt/Golden Rice; supermouse/Dolly); validation level differs (colony vs Southern-era vs founder genotyping).
 - L7.** Ti/T-DNA/borders/vir/binary + totipotency tie-in; disarmed systems noted.
 - L8.** sgRNA→PAM→R-loop→DSB→repair choice; knockout vs knock-in from repair pathway.
 - L9.** Base/prime vs nuclease: DSB-free mechanisms, bystander/off-target trade-offs, efficiency caveats.
 - L10.** Transgenesis adds vs editing changes; use-both argument (e.g., edited allele + transgenic reporter); off-target/mosaicism treatment.
 - L11.** DGE → fate/determination/differentiation/potency; master-regulator sufficiency caveat (myoD).
 - L12.** Shh source/gradient/threshold fates (V3→MN→V2→V1); duration dependence.
 - L13.** Hierarchy with cross-regulation; each layer sharpens spatial precision (Module 8 §7).
 - L14.** Enhancer (distal, orientation-flexible) vs promoter (proximal, initiation) vs silencer vs operator; experimental demos: reporter assays, ATAC/ChIP, repressor systems.
 - L15.** Motifs → behaviors with examples (toggle→borders; FFL→filtering; lateral inhibition→salt-and-pepper).
 - L16.** Method table + decision guide (Module 11 §6); full marks require "validate broadly→spatially" logic.
 - L17.** ISH workflow + probe logic + domain-vs-gradient interpretation criterion.
 - L18.** Stats on ΔC_t ; ISH spatial validation; Western/protein check; functional follow-up (e.g., morpholino/CRISPR concordance).
 - L19.** scRNA pipeline steps + claim each licenses (types; trajectories ≠ clock time).
 - L20.** Reporter types + controls + validity ceiling + endogenous validation pairing.
-

Part 4 — Viva answers (one-line keys)

- 1. Insert = DNA of interest; vector = replicating vehicle.
- 2. ori, selectable marker, (MCS/promoter for use case).
- 3. High copy amplifies toxic-gene burden → use low-copy/inducible.
- 4. 5'-AATT sticky overhangs (staggered cut).
- 5. Both leave identical 5'-GATC overhangs (compatible ends).
- 6. ATP.
- 7. Kinetics vary by ends/lengths; 3:1 is a starting heuristic.
- 8. White = lacZ α disrupted (candidate); doesn't prove insert size/sequence.
- 9. Product only forms if primer sites face each other in the correct orientation.
- 10. Two origins/markers → build in *E. coli*, propagate/use in second host (yeast/mammal).

11. Overlap-driven annealing replaces restriction-site dependence.
12. Cuts outside recognition site → overhang bases are user-chosen.
13. Correct ligation destroys recognition sites → product can't be re-cut → accumulates.
14. Verified archive of the gene flanked by att sites for transfer to destinations.
15. No useful sites anywhere on either molecule (accept inefficiency trade-off).
16. High-fidelity polymerases produce blunt ends; TA needs 3'-A.
17. Organism carrying foreign (cross-species) DNA.
18. SCNT copied an existing genome; no new DNA was designed in.
19. T-DNA cassette plasmid + helper vir functions (disarmed Agrobacterium).
20. One modified somatic cell regenerates a whole plant.
21. Required adjacent motif for Cas9 recognition/initiation.
22. NHEJ = error-prone end joining (indels); HDR = template-directed repair.
23. HDR needs donor design, cell-cycle context; efficiency low.
24. Different edits in different cells of a founder; arises when editing post-dates the first division.
25. CRISPRi represses (dCas9-KRAB); CRISPRa activates (dCas9-activators).
26. No DSB → no dependence on HDR; deaminase acts on ssDNA directly.
27. Autocatalytic chromophore; needs only O₂.
28. Fixed-tissue spatial histology; enzyme-amplified signal (X-gal).
29. Transfection-efficiency/cell-number variation between wells.
30. Degradation-tagged reporter → reports *current* expression dynamics.
31. Missing chromatin/distal context; fusion artifacts; maturation/stability kinetics.
32. Eukaryotic translation-initiation consensus context around AUG (gccRccAUGG).
33. Expression becomes spatially restricted — localization study vs production.
34. Same genome, different transcriptional outputs across cells.
35. Concentration field instructing threshold fates (Shh; Bicoid).
36. Conserved A–P identity clusters; order matches expression order (colinearity).
37. Monotonic continuous decline vs bounded step; correlation/profile criterion.
38. Dissociation removes coordinates; spatial transcriptomics (array/imaging) restores them.
39. Non-specific background (no intended pairing possible).
40. Ct values ~normal; fold changes skewed → valid statistics on ΔCt .

Part 5 — Case answers (key points)

- C1.** (a) resistance evolution under selection; (b) refuges keep susceptible mates available → dilute resistance alleles; (c) pest susceptibility bioassays; resistance-allele frequency monitoring.
- C2.** (a) *psy+crtI* complete β -carotene path in endosperm; (b) compositional equivalence, allergenicity/tox of new proteins, carotenoid bioavailability; (c) evidence→risk→ethics (autonomy, alternative interventions)→policy (approvals vary).
- C3.** Insulin: small, non-glycosylated, disulfide-processed → bacterial/yeast viable. EPO: glycosylation required for stability/half-life → CHO-class host.
- C4.** (a) epigenetic silencing; insertion position effects; allele segregation loss in breeding; (b) copy number/integration-site analysis + expression methylation assays.
- C5.** (a) models reproduce partial pathobiology; translation can fail; (b) face, construct/content, predictive validity.

C6. (a) NHEJ can be in-frame; functional domains may tolerate; (b) frameshift-allele screening (deeper genotyping); protein/function assay.

C7. (a) US product-based framing (no plant-pest DNA) → historically outside USDA; (b) EU process-based → regulated as GMO.

C8. (a) Hox genes specify segmental identity (ectopic gain-of-function transforms identity); (b) colinearity implies shared regulatory/chromatin mechanism rather than independent enhancers.

C9. (a) sufficiency of Bicoid for anterior patterning; (b) gradient forms from localized maternal mRNA → maternal supply is instructive for A–P axis.

C10. (a) morphogen gradient with threshold fates; (b) pulse/step experiments altering duration vs concentration independently (cell-culture assays/reporters).

C11. (a) dissociation discards coordinates; (b) spatial transcriptomics (spot-resolved), RNA-FISH/imaging (single-cell, subcellular localization).

C12. (a) hypothesis of ligand–receiver signaling axis; (b) validate by ISH/IF for both molecules + perturbation (ligand/receptor KO or CRISPRi) with downstream-target readout.

Part 6 — Data-question keys

D1. Y correct ($4.05+0.85 = 4.05$ kb... precisely vector 3.2 + insert 0.85); X = empty vector (single 3.2 band).

D2. Sum = 4.9 kb $\neq$ 3.2 kb → rearrangement/duplication (fragment-sum rule).

D3. Ground truth: 14 correct / 4 empty / 2 wrong-orientation / 2 mixed / 2 PCR-failure; discrepancy example: PCR-positive blue colony (lacZ α -sparing insertion) → carry to digest.

D4. Correct-clone rate L2 > L4 > L3 >> L1; colony count alone ranks L3 high while its clone quality is poor — screening rates, not counts.

D5. *myod1* 18 hpf FC $\approx 7.5\times$ ($\Delta\Delta\text{Ct} -2.9$); report ANOVA on ΔCt , $n=4$, reference = geometric mean.

D6. With unstable reference, induction is artifact; stable-reference normalization reduces/abolishes it — reference validation precedes interpretation.

D7. 18 genes at $\text{FDR}<0.05$ & $|\log_2\text{FC}|>1$; top hits form a coherent 5-gene module (per provided pathway list).

D8. $k=3$ → neural (*pax6a*) / muscle (*myod1*) / blood (*hbbe1*); $k=4$ subcluster = low-*myod1*/low-*actc1* transition state.

D9. Region markers per generator script; "notochord-like" = region whose top marker is the notochord gene with contiguous spatial domain.

D10. $r \approx -0.6$ (expression declines with distance from edge); gradient criterion = monotonic continuous decline vs bounded step; validate by ISH with sense/known-marker controls.

End of answer key. For lab-specific keys see the [Workbook](../WORKBOOK/GMO-Cloning-Gene-Expression-Lab-Workbook.md).